

CSE 420: Biomedical Image Computing and Modeling

Lung and Lung Lobe Segmentation on the LOCCA Dataset Using LobePrior, MedSAM, and MONAI

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1. Abstract

Accurate lung lobe segmentation is essential for thoracic disease analysis, particularly in pathological cases where anatomical structures are severely deformed. This study presents a comparative evaluation of three segmentation approaches on the LOCCA dataset: (1) LobePrior, a prior-driven multi-decoder architecture with anatomical template guidance; (2) MONAI 3D U-Net, a standard volumetric segmentation network; and (3) MedSAM, a foundation model adapted for medical imaging. We evaluate all three methods on identical CT volumes from COVID-19 and cancer patients, providing quantitative analysis via Dice similarity coefficient and qualitative assessment through 2D/3D visualizations. Additionally, we present a CPU-optimized implementation of LobePrior, demonstrating its feasibility for resource-constrained clinical environments. Results show that anatomical prior integration significantly improves segmentation accuracy in severely diseased lungs.

2. Introduction

2.0.1 Background: Importance of Lung Lobe Segmentation

Lung lobe segmentation plays a fundamental role in modern thoracic imaging analysis, enabling region-specific assessment of pulmonary diseases, quantitative evaluation of lesion burden, and precise anatomical localization for treatment planning. Applications such as radiation therapy, computer-aided diagnosis, surgical navigation, and longitudinal disease monitoring all rely on accurate delineation of the five pulmonary lobes. The demand for robust lobe segmentation methods has intensified in recent years due to the growing prevalence of pathologies—such as COVID-19, pneumonia, and lung cancer—that distort normal anatomy and obscure fissure boundaries. These abnormalities frequently disrupt the visual cues used by automated systems, making reliable segmentation a challenging yet essential task in clinical research and practice.

2.0.2 LOCCA Dataset: A Benchmark for Diseased Lung Segmentation

To support the development and evaluation of segmentation algorithms under such challenging conditions, the LOCCA dataset provides a high-quality, manually annotated collection of lung lobe masks in COVID-19 and cancer patients. The dataset, introduced by Ribeiro *et al.*, consists of three-dimensional CT volumes with expert delineations of all five lobes. Unlike datasets focused on healthy lungs, LOCCA captures severe pathological variations including extensive consolidations, masses, and anatomical deformations. As one of the first publicly available resources of its kind, it offers an important benchmark for assessing segmentation models in clinically relevant, disease-heavy scenarios.

2.0.3 Segmentation Methods Under Study

Several contemporary approaches have been proposed to address the inherent difficulties of lung lobe segmentation in the presence of pathology. Among them, **LobePrior** represents a prior-guided deep learning framework that integrates anatomical templates with a multi-decoder U-Net architecture. By combining coarse-to-fine processing with patient-specific prior registration, the model enhances fissure boundary consistency and preserves anatomical plausibility even when local visual features are disrupted.

In contrast, **MedSAM** extends the Segment Anything Model to medical imaging, leveraging large-scale pretraining to achieve generalizable, promptable segmentation. MedSAM performs slice-wise inference guided by bounding-box and point prompts, enabling it to adapt to diverse anatomical structures but often lacking explicit awareness of lobe-level anatomy.

Complementing these strategies, the **MONAI** ecosystem provides standardized medical imaging tools, including implementations accessible through MATLAB’s Medical Image Labeler app. Using MONAI Label, clinicians and researchers can apply pre-trained deep learning models interactively, generate automated lobe predictions, and refine outputs through manual correction—offering a flexible and reproducible workflow for segmentation tasks.

2.0.4 Study Objective and Evaluation Strategy

In this study, all three approaches—LobePrior, MedSAM, and MONAI—are applied to the same CT volume from the LOCCA dataset to enable direct comparison under identical conditions. Quantitative evaluation is performed using the Dice Similarity Coefficient for each pulmonary lobe, while qualitative analysis includes slice-based visualization and three-dimensional reconstructions. Through this combined metric and visual assessment, the strengths and limitations of each method are examined to determine which approach provides the most reliable segmentation performance in severely diseased lungs.

Contributions:

- First direct comparison of LobePrior, MONAI U-Net, and MedSAM on the LOCCA dataset
- CPU-optimized implementation of LobePrior for accessible deployment
- Quantitative and qualitative analysis on severely diseased lungs

3. Methodology

3.1 MedSAM: Promptable Foundation Model

In this work, MedSAM was used to perform lung segmentation on 3D CT volumes from the LOCCA/HCU dataset. The dataset contained volumetric CT images and their corresponding ground-truth lobe annotations in NRRD format. I first matched each CT file with its corresponding lobe label file using the case identifiers (e.g., *locca_covid_003*). After matching, I loaded the CT volumes using SimpleITK, converted them to NumPy arrays, and applied lung-specific HU windowing (−1000 to +400 HU). Each axial slice was normalized to 8-bit intensity and converted into a three-channel RGB image, as required by the MedSAM architecture.

To prepare inputs for MedSAM, I generated an automatic bounding-box prompt for every slice. This was done by thresholding the CT slice at −400 HU to isolate lung-like low-density regions, removing boundary noise, and identifying the largest connected component. The bounding box of this region was used as MedSAM’s positional prompt. For slices with minimal lung presence, I used a fallback full-slice bounding box to ensure stable inference. Because the runtime environment was CPU-only, I loaded the MedSAM checkpoint using a CPU-safe mapping procedure and executed inference sequentially for all slices in the volume.

After running MedSAM slice-wise, I combined all predicted slices to reconstruct the full 3D segmentation mask. The predicted volume was saved in NIfTI format while preserving original spatial metadata. I then evaluated the MedSAM segmentation against the ground-truth binary lung mask and the individual lobe labels using the Dice Similarity Coefficient. This provided both whole-lung Dice and lobe-wise Dice metrics, allowing me to assess the performance of MedSAM across different anatomical regions.

3.2 LobePrior: Prior-Guided Multi-Decoder Architecture

LobePrior is a deep learning method that combines convolutional neural networks with anatomical prior information for robust lung lobe segmentation in diseased lungs. Unlike standard networks relying solely on local features, LobePrior uses probabilistic anatomical

templates from reference CT scans to guide segmentation when lobar fissures are obscured by pathology.

The architecture employs a two-stage coarse-to-fine pipeline. The first stage uses a low-resolution U-Net (128^3 voxels) with a shared encoder and seven specialized decoders predicting lung parenchyma, five lobes (LUL, LLL, RUL, RML, RLL), and airways using instance normalization and self-attention modules. The second stage refines predictions via a high-resolution U-Net processing 96^3 patches with 14-channel input: original CT (1 channel), upsampled coarse predictions (7 channels), and registered anatomical template (6 channels). Sliding window inference with Gaussian weighting handles full-volume processing.

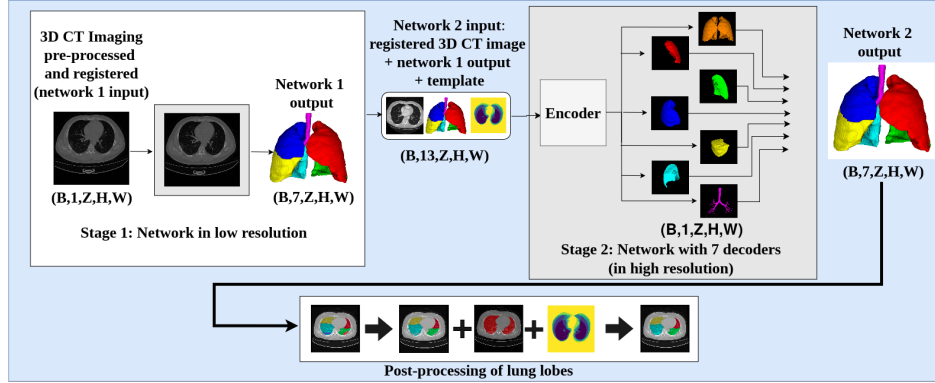


Figure 1: A diagram illustrating how the method integrates CNN-based processing with anatomical priors, which are incorporated into the network input as probabilistic models.

Anatomical priors are generated through registration-based template matching using a library of 10 reference CT scans with expert annotations. For each test image, rigid registration is performed with all references, evaluated using MSE, NCC, and MI metrics. The best-matching template is transformed to the patient’s space, providing guidance even when fissures are invisible.

The model was pre-trained on 150 CT volumes using Dice Loss and AdamW optimizer ($\text{lr}=0.0001$). We used pre-trained weights without LOCCA fine-tuning, demonstrating generalization. Our CPU-optimized implementation allocates tensors on CPU devices with batch size 1 and parallelizes registration across multiple cores using Python multiprocessing. Modified files include `predict.py`, `predict_decoders.py`, `predict_normal.py`, `predict_lung.py`, and `unet_diedre.py`. This maintains GPU-equivalent accuracy while enabling resource-constrained deployment. Inference involves HU clipping (-1024 to 600), $1 \times 1 \times 1$ mm resampling, parallel template selection, coarse-to-fine prediction, and morphological post-processing—fully automatic without manual intervention.

3.3 Lung Lobe Segmentation Using MONAI Label in MATLAB

Lung lobe segmentation was performed using the Medical Image Labeler app integrated with the MONAI Label framework. The dataset “images_HCU_part_1.zip” was obtained from the REDU repository and extracted locally. The CT volume of interest, `locca_covid_003.nii.gz`, was selected for analysis. The Medical Imaging Toolbox Interface for MONAI Label was installed through the MATLAB Add-On Explorer, enabling MONAI Label server integration within the labeling environment.

A new 3D volume labeling session was created in the Medical Image Labeler app, and the CT scan was imported into the session. The MONAI Label server was initialized by selecting “Start MONAI Label on Local Machine” from the MONAI Label tab, which triggered the automatic download of the required pre-trained segmentation models. After initialization, the segmentation model available in the MONAI Label model gallery was selected, and output classes were mapped to lobe-specific label definitions, including *lung_upper_lobe_left*, *lung_upper_lobe_right*, *lung_middle_lobe_right*, *lung_lower_lobe_left*, and *lung_lower_lobe_right*. The segmentation was then executed, producing automated lobe predictions across the CT volume. The resulting masks were reviewed slice-by-slice, and misclassified or poorly segmented regions were manually corrected using the built-in drawing and erasing tools. After refinement, the final labeled volume was exported through the “Export Ground Truth” option, generating a `groundTruthMedical` object for downstream analysis and comparison.

4. Results

4.1 Qualitative Analysis

Qualitative visualization is essential for evaluating medical image segmentation, as it reveals aspects of anatomical accuracy that numeric metrics may overlook. By inspecting representative coronal, sagittal, and transverse slices with overlaid segmentation masks, it becomes possible to assess fissure clarity, boundary continuity, and the handling of pathological regions. This visual examination provides complementary insight into the anatomical reliability and practical performance of the segmentation methods.

4.1.1 MEDSAM

MedSAM produces coarse lung masks with noticeable leakage and inconsistent boundaries across all orthogonal views due to the absence of anatomical lobe priors.

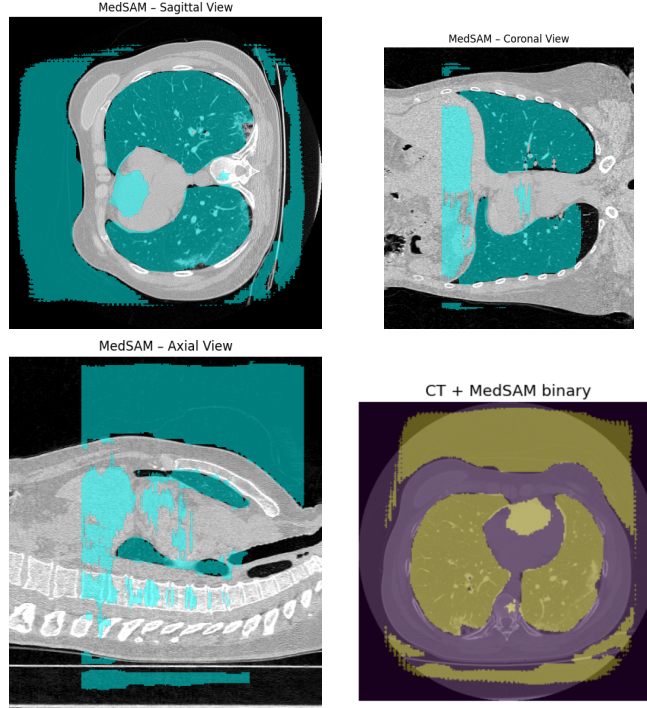


Figure 2: Lung segmentation orthogonal: coronal, sagittal,transverse and MedSAM binary.

4.1.2 LobePrior

The accurate lobe boundaries arise from LobePrior’s two-stage design, where coarse global context is refined by high-resolution decoders that integrate CT features with prior-based anatomical guidance to produce precise and anatomically consistent segmentations

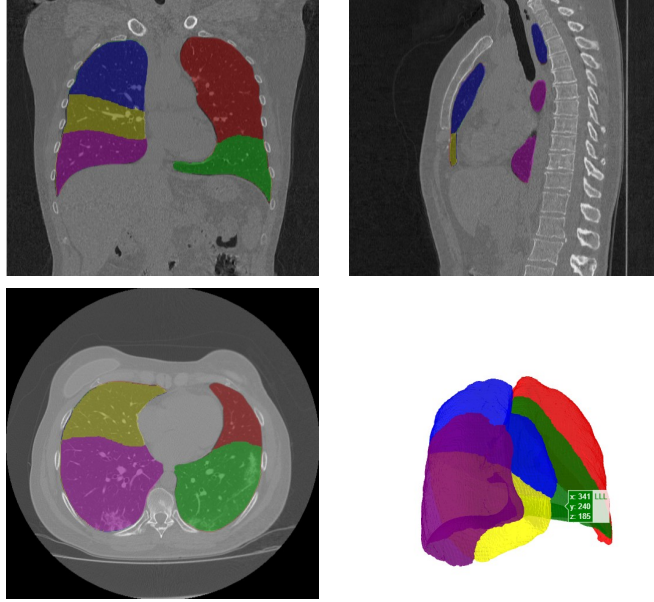


Figure 3: Orthogonal CT views: coronal, sagittal,transverse and Volume using LobePrior.

4.1.3 MONAI

Qualitative results from the MONAI segmentation demonstrate how effectively the model delineates lobe boundaries across coronal, sagittal, and transverse planes.

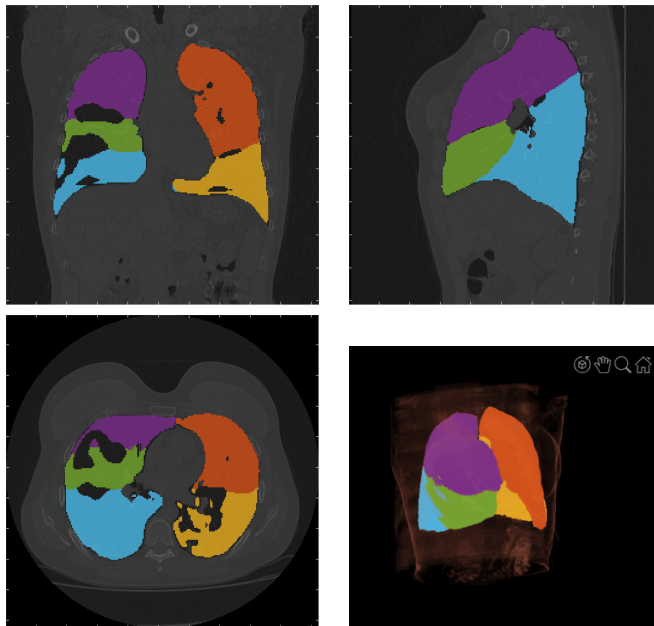


Figure 4: Orthogonal CT views: coronal, sagittal,transverse and Volume.

4.2 Quantitative Comparison

4.2.1 Evaluation Metrics

The performance of the three segmentation methods was assessed using the Dice Similarity Coefficient (DSC), a widely used overlap-based metric for evaluating volumetric segmentation accuracy. For each pulmonary lobe, the DSC quantifies the agreement between the predicted segmentation and the manually annotated ground truth in the LOCCA dataset, defined as the normalized ratio of twice the intersection over the sum of both regions. Higher Dice values indicate better correspondence and fewer boundary errors. In addition to per-lobe Dice scores, the mean Dice across all lobes was computed to evaluate overall model performance, and the standard deviation was used to characterize consistency and robustness across anatomical regions. Together, these metrics enable a comprehensive comparison of the relative strengths of LobePrior, MedSAM, and MONAI in segmenting severely diseased lungs.

Table 1: Dice Similarity Coefficient for each lobe.

Lobe	LobePrior	MONAI
LUL	0.9833	0.1713
LLL	0.9822	0.5591
RUL	0.9818	0.2093
RML	0.9678	0.3399
RLL	0.9810	0.0003
Mean $\pm$ Std	0.9792 $\pm$ 0.0057	0.2551 $\pm$ 0.1934

5. Conclusion

This study compared three approaches—LobePrior, MONAI, and MedSAM—on the LOCCA dataset, which contains highly diseased lungs with distorted fissures. The results show that only LobePrior and MONAI were able to perform true lobe-level segmentation, while MedSAM was limited to whole-lung segmentation and could not separate individual lobes.

LobePrior achieved the highest Dice scores across all lobes because its two-stage architecture combines coarse global context with fine-resolution refinement and incorporates anatomical priors. These priors allow the model to infer fissure boundaries even when pathology makes them visually unclear.

MONAI produced moderate lobe segmentation quality. Since it relies purely on image features without prior-based anatomical guidance, it struggled in areas where disease obscured the fissures.

In contrast, MedSAM—being a slice-wise, prompt-based foundation model—does not inherently model 3D lung anatomy or fissure-level structure. As a result, it successfully segmented the lungs but could not reliably separate the five lobes.

Overall, the evaluation highlights that anatomical priors and 3D contextual reasoning are critical for accurate lobe segmentation in severely diseased lungs, with LobePrior demonstrating the most robust performance.

References

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